Grade: Check Plus

STERN SCHOOL OF BUSINESS
NEW YORK UNIVERSITY

EXECUTIVE MBA
DC CLASS OF 2027

COR1.1205.85
Statistics and Data Analysis

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ASSIGNMENT 2

Course Supplement, Page 129:

EXPECTED VALUES AND OPTIONS

Due: February 24, 2026 before 6 PM

1. EXPECTED VALUES AND OPTIONS

A stock has a current selling price of $20. The distribution of the price of the stock one year from now is believed to be as follows:

PRICE (x)	PROBABILITY f(x)
20	.05
21	.06
22	.07
23	.08
24	.09
25	.12
26	.15
27	.13
28	.10
29	.10
30	.05
	1.00

An investor holds a one year option (European) on this stock. This enables him to purchase, one year from now, one share of stock at $25 per share. The value of his option will depend upon the price of the stock one year from now. If the price rises to $28, the value of the option (ignoring interest and commissions) is $3. If the price rises to only $23, the option is worthless.

1. Compute the mean and standard deviation of the stock price.
2. a. What is the probability that the stock price will fall within 1 standard deviation of the mean one year from now?
   b. What is the probability that the stock price will fall within 2 standard deviations of the mean one year from now?
3. Derive the distribution of the option value, ignoring interest and commissions.
4. What is the probability that the option will have positive value?
5. Compute the mean and standard deviation of the value of the option.

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Assignment 2 – James Yang

price(x)	probability f(x)	$x \cdot f(x)$	$(x-\mu)$	$(x-\mu)^2$	$x-\mu^2 \times f(x)$
20	0.05	1	-5.4	29.2681	1.4634
21	0.06	1.26	-4.41	19.4481	1.1669
22	0.07	1.54	-3.41	11.6281	0.8140
23	0.08	1.84	-2.41	5.8081	0.4646
24	0.09	2.16	-1.41	1.9881	0.1789
25	0.12	3	-0.41	0.1681	0.0202
26	0.15	3.9	0.59	0.3481	0.0522
27	0.13	3.51	1.59	2.5281	0.3287
28	0.10	2.8	2.59	6.7081	0.6708
29	0.10	2.9	3.59	12.8881	1.2888
30	0.05	1.5	4.59	21.0681	1.053

$\mu = 25.41$ $\sigma^2 = 7.5015$

1a. $\mu = \sum_{\text{all } x} f(x) = 1 + 1.26 + 1.54 + 1.84 + 2.16 + 3 + 3.9 + 3.51 + 2.8 + 2.9 + 1.5$ ✓ Q1

$\mu = 25.41$ – mean or expected value

$\sigma = \sqrt{\sum_{\text{all } x} (x-\mu)^2 \cdot f(x)}$

$\sigma = \sqrt{1.4634 + 1.1669 + 0.8140 + 0.4646 + .1789 + .0202 + .0522 + .3287 + .6708 + 1.2888 + 1.053}$

cont'd

$\sigma = \sqrt{7.5015}$

$\sigma = 2.7389$ – standard deviation

2a. $P(\mu - \sigma \le X \le \mu + \sigma) = P(25.41 - 2.74 \le 25.41 + 2.74)$ ✓ Q2a

$P(22.67 \le X \le 28.15)$ whole numbers so...

$P(23 \le X \le 28) = 0.08 + 0.09 + 0.12 + 0.15 + 0.13 + 0.10$

$= 0.67$ or 67% ✓ Q2b

2b. $P(\mu - 2\sigma \le X \le \mu + 2\sigma) = (25.41 - 5.48 \le X \le 25.41 + 5.48)$

$P(19.93 \le X \le 30.89)$ whole numbers so...

$P(20 \le X \le 30)$ = all possible stock prices so 0.1 or 100%

3. value of option

X	option value (Q)	prob. or f(Q)	option value · f(Q)
20-25	0	0.47	0
26	1	0.15	0.15
27	2	0.13	0.26
28	3	0.10	0.3
29	4	0.10	0.4
30	5	0.05	0.25

μ(option) = 1.36

Column 4 is distribution of option value.

✓ Q3

4. Probability that option will have positive value

✓ Q4

$= P(x=26 + x=27 + x=28 + x=29 + x=30)$

$= 0.15 + 0.13 + 0.10 + 0.10 + 0.05 = 0.53$ or 53%

5. mean of value of option (Q)

$\mu = \sum_{\text{all } Q} \cdot f(Q) = 0 + 0.15 + 0.26 + 0.3 + 0.4 + 0.25 = 1.36$

μ of option value = 1.36

standard deviation of Q

option value Q	f(Q)	Q-1.36	$(Q-1.36)^2$	$(Q-1.36)^2 \cdot f(x)$
0	0.47	-1.36	1.8496	0.869312
1	0.15	-0.36	0.1296	0.01944
2	0.13	0.64	0.4096	0.053248
3	0.10	1.64	2.6896	0.26896
4	0.10	2.64	6.9696	0.69696
5	0.05	3.64	13.2496	0.66248

✓ Q5

$\sigma^2 = \sum_{\text{all } Q} (Q-\mu)^2 \cdot f(Q)$ = sum of fourth column

$= 2.5724$ $\quad \sigma = \sqrt{\sigma^2} = \sqrt{2.5724}$

$\sigma = 1.6038$